

YOUSSEF YASSER ELSAYED HAMED

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PROFILE

Third-year CS and AI student at Cairo University specializing in Machine Learning, Deep Learning, and Computer Vision, with hands-on experience building end-to-end ML systems for real-world applications. Strong in problem-solving, model optimization, and developing reliable AI solutions that are practical, scalable, and production-ready.

EDUCATION

Cairo University

Bachelor's in Computer Science and Artificial Intelligence

Giza, Egypt

Sep. 2023 – Sep. 2027

EXPERIENCE

AI Engineer Intern

Novus

April 2026 – Present

On-site, New Cairo, Egypt

- Engineered a real-time **Behavioral AI Theft Detection** pipeline using **YOLOv11** + **ByteTrack** + **LSTM** to classify suspicious human-object interactions across 30 high-value target classes in retail environments.
- Maintained production-level code quality through modular architecture, peer code reviews, and iterative model improvements within a professional on-site engineering team.

Machine Learning Intern

Code Alpha

April 2026

Remote

- Delivered two production-ready classifiers under real deadlines — a **47-class** handwritten character recognizer (131K+ images, **94%** accuracy) and a binary tumor malignancy detector (560+ instances, **96%** accuracy).

Competitive Programmer

Egyptian Collegiate Programming Contest (ECPC)

July 2024

Alexandria, Egypt

- Engineered optimized algorithmic solutions in **C++** under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

PROJECTS

YAI — Video Safety Analyzer [GitHub] | [Live Demo](#) | *PyTorch FastAPI, YOLOv10x, CLIP, ByteTrack* May 2026

- End-to-End CV Pipeline:** Engineered a full-stack real-time video safety analysis system with a **FastAPI** backend and **Netlify**-hosted frontend, processing uploaded footage through a **5-stage** inference pipeline — detection, classification, tracking, anomaly scoring, and depth estimation — returning a structured JSON report .
- Multi-Model Inference Stack:** Integrated **YOLOv10x** + **CLIP** open-vocabulary classification for frame-level detection, **ByteTrack** + **Kalman filtering** for persistent multi-person tracking across occlusions, **AnomalyTransformer** for behavioral anomaly scoring across **6** threat categories (loitering, crowd surge, forbidden zone, speed anomaly, trajectory anomaly, disappearance), and **Depth Anything v2** for monocular depth estimation — all running on a **CUDA T4 GPU**.
- Production-Grade Backend:** Built an async **FastAPI** server with **WebSocket** streaming for real-time frame-level result delivery, job queue management, and a **REST** polling API — tunneled via a permanent **ngrok** domain with **zero-downtime** frontend decoupling between Netlify and the GPU backend.

Adaptive Pose & Action Intelligence System [GitHub] | *Python, PyTorch, YOLOv8, OpenCV, SQLite* May 2026

- Multi-Level Human Analysis Pipeline:** Engineered a real-time computer vision system simultaneously tracking **527** anatomical points across 3 analysis levels: **17** body keypoints via YOLOv8-Pose, hand state estimation, and **68** facial landmarks via YuNet + LBF — achieving **55 FPS** .
- Action & Posture Intelligence:** Extracted **8** joint angles per frame to compute real-time posture scores (0–100) and classify **6** body actions using a hybrid rule-based + PyTorch **LSTM** pipeline operating on 30-frame temporal sequences.
- 3-Layer Memory System:** Implemented a persistent behavioral memory using short-term RAM deque, unsupervised pattern learning across sessions, and a **SQLite** long-term store — enabling cross-session posture trend tracking and anomaly flagging with threaded face analysis for optimal throughput.

Behavioral AI Theft Detection System [GitHub] | *Python, PyTorch, YOLOv11, SAHI, OpenCV*

April 2026

- **Real-Time Vision Pipeline:** Engineered a computer vision pipeline using **YOLOv11x** for high-fidelity pose estimation and instance segmentation, actively monitoring **30** high-value target classes.
 - **Robust Detection & Tracking:** Integrated **SAHI** (1000×1000 tiling) to maximize small-object recall, with **ByteTrack** and a **ResNet18**-based ReID manager for persistent multi-target tracking across occlusions.
 - **Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train a custom PyTorch **LSTM** network with an attention mechanism for classifying suspicious temporal movements.
- Plant Disease Recognition System** [Kaggle] | *Python, TensorFlow/Keras, CNN, OpenCV* March 2026
- **Automated Diagnosis:** Developed a deep learning system classifying **38** distinct plant disease categories using the PlantVillage dataset using **54,000+** images, achieving **99%** validation accuracy.
 - **Model Architecture & Training:** Designed and optimized a **Convolutional Neural Network (CNN)** pipeline for high-resolution image classification, efficiently analyzing over **54,000** images across various species.
 - **Comprehensive Evaluation:** Validated model performance using Confusion Matrices, Precision-Recall metrics, and per-class accuracy analysis across all 38 disease categories.
- Traffic Sign Image Classifier** [GitHub] | *Python, PCA, KNN, Naive Bayes, MobileNet V2* Feb. – March 2026
- **Traffic Sign Classification:** Developed a traffic sign classification system for **43** classes using **39,000+** images, achieving **98%** accuracy in road sign categories using the GTSRB dataset.
 - **Model Comparison:** Benchmarked classical ML baselines (KNN, Naive Bayes) against a fine-tuned **MobileNet V2** CNN, boosting final accuracy to **98%**.
- Student Performance Prediction System** [Kaggle] | *Random Forest, Gradient Boosting, SHAP, Pandas* Dec. 2025
- **Full ML Pipeline:** Binary classifier predicting student pass/fail over **1,000+** records across **10** stages — EDA, feature engineering, preprocessing, training, tuning, and SHAP interpretability .
 - **Model Comparison:** Benchmarked **5** classifiers (Random Forest, Gradient Boosting, SVM, Logistic Regression, Decision Tree); Random Forest ranked **#1** — tuned via **GridSearchCV** + **Stratified K-Fold**, achieving **91%+** accuracy and top ROC-AUC on held-out test set.
 - **Feature Engineering & Interpretability:** Engineered **2** high-signal features (avg_score, edu_level); applied **SHAP TreeExplainer** to rank feature contributions — identifying test preparation and lunch type as the **top 2** behavioral predictors.

CERTIFICATES & AWARDS

Deep Learning Specialization <i>DeepLearning.AI (Andrew Ng) – Coursera</i>	Feb. 2026
Intermediate Machine Learning <i>Kaggle</i>	March 2026
Intro to Deep Learning <i>NVIDIA</i>	Dec. 2025
Egyptian Collegiate Programming Contest (ECPC) Participant	July 2024

TECHNICAL SKILLS

Programming & Tools: Python, C++, Git, GitHub, Jupyter Notebook, VS Code, Kaggle, SQLite

Machine Learning & AI: Deep Learning, Computer Vision, Predictive Modeling, CNNs, Object Detection, Pose Estimation, Instance Segmentation, Anomaly Detection, Depth Estimation, Action Recognition, Behavioral Analysis, Transfer Learning, Hyperparameter Tuning, Random Forest, KNN, Naive Bayes, Logistic Regression, LSTM, Attention Mechanisms, Transformer Models

Computer Vision & Perception: Multi-Object Tracking, Re-Identification (ReID), Optical Flow, Monocular Depth Estimation, Facial Landmark Detection, Hand State Estimation, Skeletal Keypoint Extraction, Open-Vocabulary Classification, Small-Object Detection (SAHI)

Data Science & Analysis: Exploratory Data Analysis (EDA), Feature Engineering, Dimensionality Reduction (PCA), Data Cleaning, Data Visualization, Data Augmentation, Statistical Analysis, Model Evaluation (ROC-AUC, Precision-Recall, Confusion Matrix, F1-Score)

Libraries & Frameworks: PyTorch, TensorFlow/Keras, Scikit-Learn, OpenCV, FastAPI, WebSocket, YOLO (v8, v10x, v11), CLIP, ByteTrack, Kalman Filter, SAHI, MediaPipe, MobileNet V2, ResNet18, Depth Anything v2, AnomalyTransformer, Pandas, NumPy, Matplotlib, Seaborn

Deployment & Engineering: REST API Design, WebSocket Streaming, Async Backend Development, GPU Inference (CUDA), ngrok Tunneling, Netlify, Vercel App, Job Queue Management, End-to-End ML System Design, Model Optimization, Pipeline Architecture

Software Engineering: Data Structures & Algorithms, Competitive Programming, System Optimization, Object-Oriented Programming, Multithreading

Soft Skills: Problem-Solving, Critical Thinking, Team Collaboration, Time Management, Adaptability, Professional Communication

LANGUAGES

Arabic: Native or bilingual proficiency
English: Professional working proficiency